



AUT@MIATA

Technology Services LLC

macOS Management

Introduction to macOS

Practical Exercise

Advanced macOS

Practical Exercise

Managing Hidden Files

Practical Exercise

macOS System Settings

Practical Exercise

macOS Utilities

Practical Exercise

Introduction to Terminal

Practical Exercise

macOS Software Update

Practical Exercise

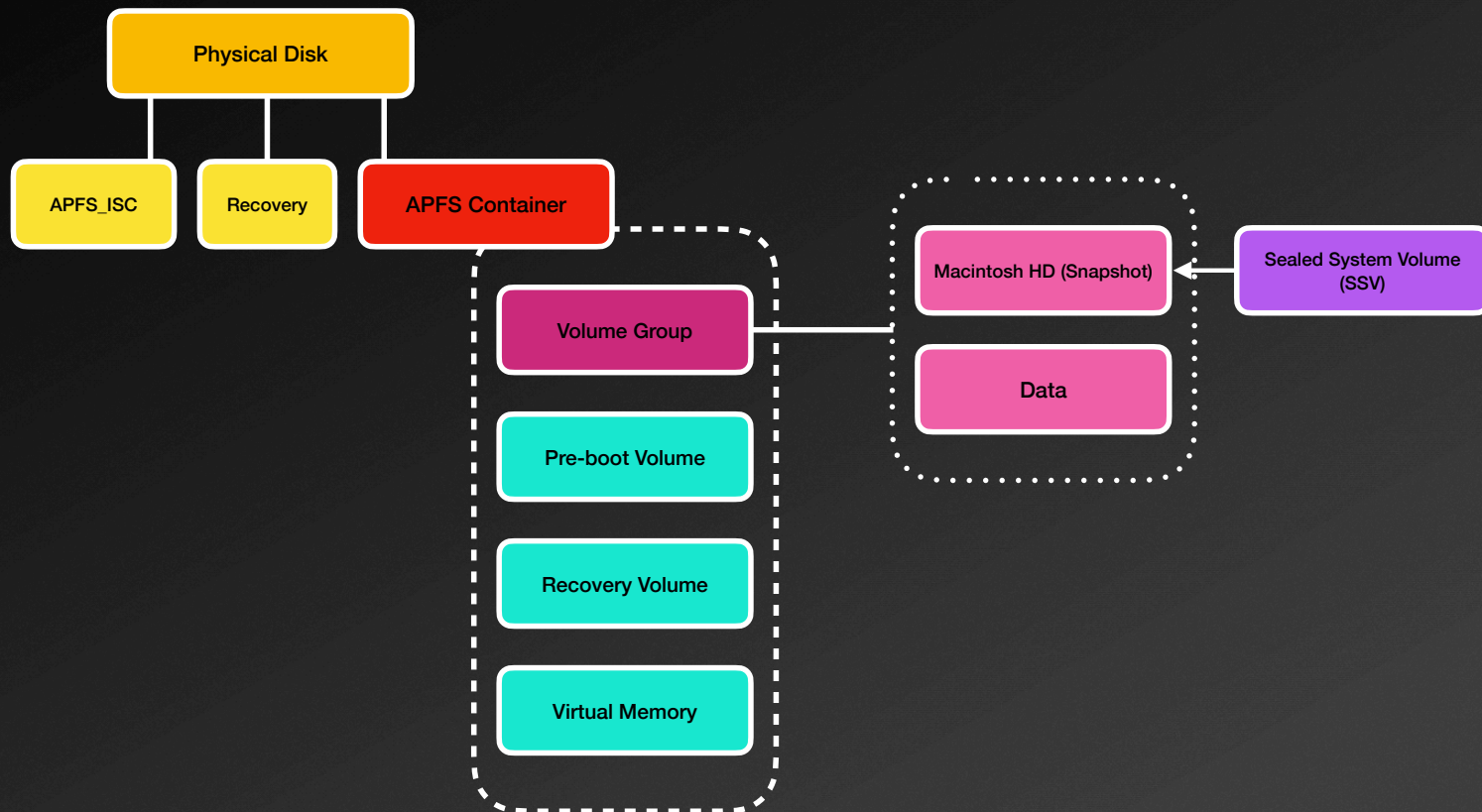
Apple File System

Practical Exercise

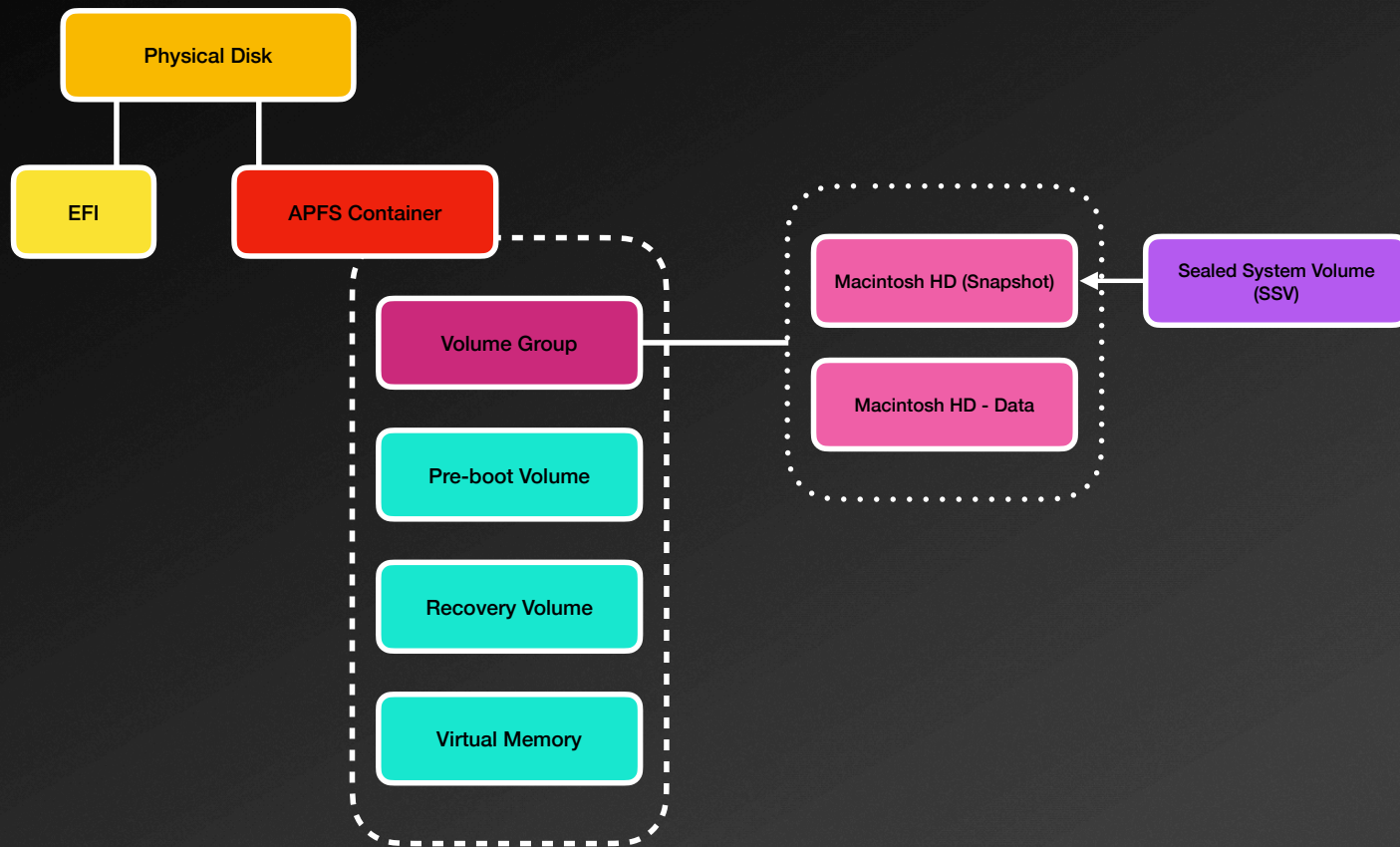
APFS vs HFS+

- HFS+ is the legacy file system for older Macs prior to macOS Sierra (10.12)
- APFS is the standard file system for current versions of macOS. Modern and integrates support for virtual volumes, snapshots, and encryption.
- APFS specifically designed for Solid State Disk style drives.
- Regardless of format, Apple uses GUID for the Partition Map for bootable macOS Media (USB, Internal Disks, etc).

APFS - Apple Silicon



APFS - Intel



Managing Mac Computer Names

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macOS Erase Assistant

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Managing User Accounts

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Directory Services Overview

- **Local User Account Scope**

- Local User Account Resident on Device

- **Network User Account**

- User Account from a Network-based LDAP like Active Directory
- Requires Mac being 'bound' to the LDAP and Connected to the Network

- **Kerberos Authentication**

- Uses Kerberos Extension to Allow Local User Account to Authenticate to Network-based LDAP without Binding.

- **Platform Single-Sign-on**

- Local User Account that syncs with an Internet-based Identity Provider like Microsoft Azure / Entra ID.

Login Items & Background Apps

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Troubleshooting WiFi

Practical Exercise

File Permissions

Practical Exercise

POSIX

d r w x r w x r w x

The diagram illustrates the POSIX permission string **drwxrwxrwx**. It features four horizontal bars of different colors: red, green, teal, and blue. A red bracket is positioned above the first character 'd'. A green T-shaped bracket is positioned above the next three characters 'rwx'. A teal T-shaped bracket is positioned above the next three characters 'rwx'. A blue T-shaped bracket is positioned above the final three characters 'rwx'. Below each of these brackets is a label: 'directory' in red, 'owner' in green, 'group' in teal, and 'everyone' in blue.

directory

owner

group

everyone

CLI Permissions Values

Value	Permission	Directory Listing
0	no read, no write, no execute	-
1	no read, no write, execute only	-x
2	no read, write only, no execute	-w-
3	no read, write & execute only	-wx
4	read only, no write, no execute	r-
5	read and execute only, no write	r-x
6	read and write only, no execute	rw-
7	read, write, and execute	rwX

macOS Security & Encryption

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Gatekeeper & Notarization

- Determines if applications or processes can run.
- Requires code signing by an Apple Developer account.
- If malware is using a signing certificate, Apple can remotely update Gatekeeper to no longer trust that certificate and keep that malware from running.
- Users will get a trust warning and must override Gatekeeper to launch an unsigned application.

XProtect

- Built-in Anti-Virus and Anti-Malware software.
- Runs in the background and Apple updates definitions regularly.
- Has the ability to identify and then clean/remove/delete known viruses and malware.

System Integrity Protection (SIP)

- Ensures that specific Unix system folders cannot be modified by 3rd party software.
- Previous to Mac OS X 10.11 (El Capitan) the root user had full control to all folders.
- Folders protected using SIP include
 - /System
 - /usr
 - /bin
 - /sbin
 - /var

Kernel vs System Extensions



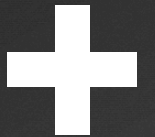
kexts



system extensions



kexts



system extensions

kexts
(reduced security req)

kexts NOT recommended!

FileVault

- Data encryption for Mac computers (similar to BitLocker on Windows)
- Encrypts Data at Rest - Full Disk Encryption
- Enforces -
 - Requires a password to access encrypted data on disks.
 - Requires a password to login/unlock/wake the Mac & access macOS Recovery.

macOS Privacy Settings

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Mac Network Authentication

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Connecting to a Windows File-Print Server

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